

P3641R0

Rename `std::observable` to `std::observable_checkpoint`, and add a feature-test macro

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<https://eisenwave.github.io/cpp-proposals/observable-checkpoint.html>

Author:

[Jan Schultke](#)

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Abstract

Rename `std::observable` to `std::observable_checkpoint`, and add a feature-test macro.

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§ 1. Introduction

[P1494R5] "Partial program correctness" introduced so-called *observable checkpoints* into the language, as well as the function `std::observable` to establish such a checkpoint in the program.

Three issues with the name `std::observable` have been brought up ([Reflector], [Reddit]):

- It doesn't express intent very well, and a longer name could do that better.
- Some associate it with the "observer" design pattern, but it is unrelated.
- It is a short name, and using it for observable checkpoints now may impede future standardization.

§ 1.1. How did it end up being called `std::observable`?

The name originated in P1494R0.

LEWG last discussed the name during its review of P1494R2 ([TeleconMinutes]). Alternative names (including `std::observable_checkpoint`) came up during that discussion, but LEWG did not poll names. The minutes of LEWG's review of P1494R3 do not record a discussion of names.

NOTE: This does not mean to imply that LEWG neglected its duties. [P1494R5] itself does not discuss alternative names or request naming feedback.

EWG reviews did not poll names, and their minutes do not record a discussion of names. No further discussion of names occurred until the plenary vote at Hagenberg 2025.

Considering the situation, a name change should be polled before Sofia 2025. This needs to be done via paper or national body comment because [P1494R5] has already been approved in plenary.

§ 2. Design

In discussions, `std::observable_checkpoint` has emerged as a favorite suggestion. It is a very good name because it is self-explanatory:

`std::observable_checkpoint()` establishes an observable checkpoint.

This expresses intent as elegantly as `std::unreachable`, which documents that a point in the program is unreachable (or else ...).

NOTE: Various other options such as `std::checkpoint` have been discussed. `std::observable_checkpoint` is proposed because it's most likely to achieve consensus.

`std::observable` on its own makes little sense because it's not trying to say that a point in the program *is observable*, or that it *is observable behavior*. Instead, `std::observable` *is a checkpoint* for prior observable behavior. This intent should be expressed in the function name.

Furthermore it has been suggested in [\[Discussion\]](#) to add a feature-test macro for the library function. After all,

- there is a `__cpp_lib_unreachable` macro, and `std::unreachable` and `std::observable` are similar,
- [\[P1494R5\]](#) adds no feature-test macro for observable checkpoints (and it shouldn't), and
- it is useful to know whether the `std::observable_checkpoint` function exists.

NOTE: I was unable to find any discussion of a feature-test macro in meeting minutes, so lack thereof may be an oversight, rather than conscious design.

§ 3. Proposed wording

The following changes are relative to [\[N5001\]](#) with the changes in [\[P1494R5\]](#) applied.

In subclause [\[intro.abstract\]](#), modify the note attached to paragraph 5 as follows:

[*Note:* A call to `std::observable_checkpoint` ([utility.undefined]) is an observable checkpoint. — *end note*]

In subclause [\[utility.syn\]](#), modify the synopsis as follows:

```
// [utility.undefined], undefined behavior  
[[noreturn]] void unreachable() noexcept;  
void observable_checkpoint() noexcept;
```

In subclause [\[version.syn\]](#), add a feature-test macro as follows:

```
#define __cpp_lib_observable_checkpoint 2025XXL // freestanding, also in
```

In subclause [\[utility.undefined\]](#), modify the declaration of `std::observable()` as follows:

```
void observable_checkpoint() noexcept;
```

Effects: Establishes an observable checkpoint ([intro.abstract]).

§ 4. Acknowledgements

Thanks to Mark Hoemmen for researching [§ 1.1 How did it end up being called std::observable?](#).

§ References

§ Normative References

[N5001]

Thomas Köppe. *Working Draft, Programming Languages — C++*. 17 December 2024. URL: <https://wg21.link/n5001>

[P1494R5]

S. David Herring. *Partial program correctness*. URL: <https://wg21.link/P1494R5>

§ Informative References

[Discussion]

[std-discussion] Should there be a library feature-test macro for std::observable?. URL: <https://lists.isocpp.org/std-discussion/2025/02/2901.php>

[Reddit]

Redditors. *WTF std::observable is?*. URL:

https://www.reddit.com/r/cpp/comments/1is7aqy/wtf_stdobservable_is/

[Reflector]

WG21. *std::observable should have a better - ie worse - name*. URL: <https://lists.isocpp.org/lib-ext/2025/02/29541.php>

[TeleconMinutes]

LEWG. *LEWG Telecon Minutes for P1494R2*. 1 February 2022. URL:

<https://wiki.edg.com/bin/view/Wg21telecons2022/P1494>